

$$P(\text{Red} \mid \text{Poisonous}) \neq P(\text{Poisonous} \mid \text{Red})$$

$$P(\text{Positive} \mid \text{Buyer}) = 0.5$$

$P(\text{Buyer} \mid \text{Positive})$

$$\mathbf{ALL} = \mathbf{TP} + \mathbf{FP} + \mathbf{FN} + \mathbf{TN}$$

$$\text{Prevalence} = \frac{\text{TP} + \text{FN}}{\text{ALL}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$P(\text{Positive}) = \frac{\text{TP} + \text{FP}}{\text{ALL}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\frac{\text{Recall}}{\text{Precision}} = \frac{\text{TP}/(\text{TP} + \text{FN})}{\text{TP}/(\text{TP} + \text{FP})} = \frac{\text{TP} + \text{FP}}{\text{TP} + \text{FN}}$$

$$\frac{\text{Recall}}{\text{Precision}} = \frac{(\text{TP} + \text{FP}) / \text{ALL}}{(\text{TP} + \text{FN}) / \text{ALL}} = \frac{P(\text{Positive})}{\text{Prevalence}}$$

$$\text{Precision} = \frac{\text{Recall} \cdot \text{Prevalence}}{P(\text{Positive})}$$

$$P(\text{Buyer} \mid \text{Positive}) = \frac{P(\text{Positive} \mid \text{Buyer}) \cdot P(\text{Buyer})}{P(\text{Positive})}$$